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Robust Nonparametric Early Stopping in Tree Ensembles via IQR-Scale Change-Point Detection

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Abstract

Tree ensembles—Random Forests (RFs) and Gradient Boosting Machines (GBMs)—often stabilize before all trees are evaluated. We study early stopping as a nonparametric change-point problem on prediction increments. The P^2-STOP method family monitors a robust interquartile-range (IQR) scale of prediction increments online and stops when a relative-scale criterion is met. The default variant uses a rolling-window exact-quantile estimator ($O(w)$ memory), which provides a clean finite-sample stopping guarantee; a full-prefix P^2 streaming approximation ($O(1)$ memory) is available as a memory-light alternative. The stopping rule applies to both RFs and GBMs without model-specific distributional assumptions. On four RF benchmarks (MNIST, Covertype, HIGGS, and Credit Card Fraud), P^2-STOP achieves 44.8% mean work reduction (range: 0.7–71.7%) with an accuracy change from -0.53 to $+0.02$ percentage points versus full-ensemble inference. On XGBoost ($T = 500$), work reduction is dataset-dependent (41.4% on Covertype up to 89.0% on Credit Card), with corresponding accuracy trade-offs. Under random-tree contamination conditions (5%, 15%, and 25%), performance remains stable, whereas IQR-versus-standard-deviation baseline differences are mixed rather than uniformly dominant. Designed for compiled inference engines (e.g., C++/Numba), P^2-STOP translates theoretical work reduction into consistent wall-clock speedups (4.14×–4.82× versus compiled full RF on MNIST/Covertype/HIGGS for $T = 500$). Native Python implementations serve purely as logical baselines due to loop overhead, while Credit Card exhibits the expected slowdown when work reduction is near zero. All comparisons use five seeds with 95% confidence intervals and seed-level paired tests. With only five seeds, inferential power is limited, and p-values should be interpreted cautiously. Relative to the Dirichlet RF baseline, our contribution is not larger RF-specific work reduction; it is a robust nonparametric IQR-scale stopping criterion, cast as a change-point/sequential-inference problem, that works as a post hoc wrapper across RF and GBM settings.

Keywords: nonparametric estimation; change-point detection; tree ensembles; streaming algorithms; quantile approximation; robust statistics; model-agnostic inference; sequential stopping

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1. Introduction

1.1. Computational Waste in Tree Ensembles

Modern tree ensembles are widely used for tabular machine learning. Random Forests [1] and Gradient Boosting Machines [2] routinely employ $T \in [100, 10{,}000]$ base

learners. Full inference requires traversing every tree for every instance, incurring $O(T)$ cost per prediction. In practice, this often creates substantial computational waste: many late trees contribute little marginal information but are still evaluated.

A crucial empirical observation motivates our work: for a large fraction of instances, the cumulative prediction stabilizes long before all T trees have been evaluated. The later trees contribute diminishing, near-redundant information. If this convergence point can be detected during inference, one can stop early and reduce evaluated trees without sacrificing accuracy.

Accordingly, we frame early stopping as a resource-budgeted inference problem. Our primary empirical targets are accuracy preservation and work reduction (fraction of trees evaluated). Because P^2-STOP relies on an $O(1)$ or $O(w)$ online stopping loop, it is inherently designed for compiled inference engines (e.g., C++ and Numba) where tree traversal and stopping logic execute seamlessly. While we report native Python results purely as a logical baseline to quantify tree-count reduction, our Numba-compiled implementation demonstrates the true end-to-end wall-clock speedup attainable under target deployment conditions.

1.2. Existing Early-Stopping Approaches and Their Limitations

Several strategies for reducing ensemble evaluation cost have been proposed:

- Budget-constrained ensemble pruning. Nan et al. [3] studied RF pruning with prediction budgets, involving learning to select subsets of trees to evaluate. Their approach is specific to RF classification and requires a training-time pruning step.
- Cost-sensitive classifier cascades. Xu et al. [4] built cost-sensitive trees of classifiers that route instances through increasingly complex models. Trapeznikov and Saligrama [5] studied budgeted sequential classification with cascaded classifiers of increasing cost. Both approaches require model modification or joint training.
- Model-agnostic ensemble fast evaluation. Wang et al. [6] proposed QWYC ("Quit When You Can"), which combines two ingredients: (i) an optimized evaluation ordering of base models and (ii) a margin-based early-stopping criterion. QWYC applies to both independently trained (e.g., RF) and sequentially trained (e.g., boosted) ensembles and is the closest prior work to ours in scope. We implement and evaluate both components: component (ii) alone with the natural tree order, and the full method (i) + (ii) with validation-optimized ordering (Section 7.1.6).

Existing approaches either require model modification [4,5], are specific to a single ensemble type [3], or rely on ordering optimization with margin-based thresholds [6]. Our work contributes a distinct robust nonparametric stopping criterion based on IQR-scale change-point detection in the prediction-increment process. Unlike QWYC's margin rule, which certifies that the leading class cannot be overtaken by the remaining trees, P^2-STOP monitors the variability structure of prediction increments directly and provides a finite-sample concentration bound for the rolling-window variant.

1.3. Our Contribution

We observe that the prediction trajectory $\{\hat{y}_1(x), \hat{y}_2(x), \ldots, \hat{y}_T(x)\}$ is a discrete-time stochastic process whose increments undergo a phase transition: an initial exploration phase of high-variance increments followed by a convergence phase of near-zero increments. This transition is precisely a change-point in the scale structure of the increment process—a classical object in mathematical statistics [7,8]. Deciding when to stop accumulating trees is, at its core, an optimal stopping problem [9,10]: one seeks to minimize computation while controlling the risk of premature termination.

Building on the P^2 (piecewise-parabolic) streaming quantile algorithm [11] and the FORCE robust scale paradigm [12], we develop the P^2-STOP method family (Section 4.1): a nonparametric change-point stopping rule that monitors the IQR-based scale of prediction increments and fires when the scale drops below a relative threshold. The family comprises two backends:

- A rolling-window exact-quantile variant ($O(w)$ memory): the default in all experiments, which provides a clean finite-sample guarantee (Theorem 1).
- A full-prefix P^2 streaming variant ($O(1)$ memory): a memory-light approximation retained as a baseline.

The name P^2-STOP reflects the intellectual lineage from the P^2 algorithm, not a commitment to the full-prefix variant as the default implementation.

We evaluated the method with seed-level paired analyses against full-ensemble inference and against Dirichlet RF stopping, emphasizing robustness and generality rather than claiming superior work reduction against Dirichlet.

Our contributions are as follows:

1. Formulation. We formulate ensemble early stopping as a nonparametric change-point detection problem in the prediction-increment process.
2. P^2-STOP algorithm. We propose a robust IQR-scale monitoring rule with fixed-size state and lightweight per-tree updates, applicable without model-specific distributional assumptions. The rolling-window exact variant is the recommended default; the full-prefix P^2 variant serves as an $O(1)$-memory alternative.
3. Robustness. The IQR-scale functional has a 25% breakdown point in the Hampel sense [13]; our contamination experiments provide empirical evidence that this robustness transfers to stable stopping behavior, though a formal breakdown-point proof for the composite sequential rule is not established here.
4. Statistical evaluation protocol. We report 95% confidence intervals and seed-level paired tests for accuracy/work trade-offs and for end-to-end inference timing.

Section 2 reviews related work. Section 3 introduces the P^2 algorithm, FORCE equations, and ensemble trajectory definitions. Section 4 presents P^2-STOP. Section 5 develops the theoretical analysis. Section 6 describes the experimental setup. Section 7 presents the empirical results. Section 8 discusses the implications and limitations. Section 9 concludes.

2. Related Work

2.1. Adaptive Computation and Early-Exit Inference

The idea of reducing inference cost by terminating computation early has been explored in several machine learning paradigms. Cascaded classifiers [5] arrange classifiers of increasing cost and stop when confidence is high. Budgeted prediction methods [3,4] learn to prune trees or route instances through subsets of an ensemble under cost constraints. Wang et al. [6] propose QWYC, which combines optimized evaluation ordering with a margin-based early-stopping criterion for general classifier ensembles including both RFs and boosted trees; our experiments evaluated QWYC in both a margin-only configuration and the full ordering-optimized configuration (Section 7.1.6). Early-exit neural networks [14] attach auxiliary classifiers to intermediate layers and exit when certainty exceeds a threshold. More recent work on dynamic networks [15,16] learns instance-dependent routing or layer-skipping policies, while adaptive computation time [17] and AdaNet [18] adapt network depth or structure during inference. All these approaches require model modification or retraining. Our work differs in that P^2-STOP operates as a pure post hoc inference wrapper: it requires no changes to the trained model.

2.2. Streaming Quantile Algorithms

Streaming quantile estimation has a rich history. The P^2 algorithm [11] uses five markers with $O(1)$ memory. Greenwald and Khanna [19] provide an ε-approximate quantile summary with $O((1/\varepsilon)\log(\varepsilon N))$ space. The t-digest [20] adaptively clusters data for accurate tail quantiles. We adopt P^2 because its strict $O(1)$ memory and $O(1)$ update cost are ideal for per-instance streaming during ensemble inference, where $N = T$ is typically 10^2–10^4 and the constant overhead must be negligible.

2.3. Sequential Analysis and Change-Point Detection

Sequential hypothesis testing originates with Wald's SPRT [10,21], which established the theoretical basis for making accept/reject decisions while controlling error probabilities as data arrive. Shiryaev extended this framework to quickest change-point detection [22], deriving Bayes-optimal stopping rules for detecting distributional shifts in sequential observations. Lorden [23] introduced the complementary minimax formulation, seeking procedures that minimize worst-case detection delay. Tartakovsky et al. [24] provide a modern treatment unifying hypothesis testing and changepoint detection; Lai [25] surveyed the broader landscape of sequential analysis from classical SPRT to contemporary applications.

Classical nonparametric methods detect distribution shifts in sequential data. The CUSUM statistic [26] monitors cumulative sums for shift detection; Moustakides [27] established its optimality for Lorden's minimax criterion. Brodsky and Darkhovsky [7] developed nonparametric tests based on rank statistics. Csörgő and Horváth [8] provide asymptotic theory. Modern kernel-based approaches [28] extend detection to multivariate settings, and computationally efficient methods such as PELT [29] enable changepoint detection with linear cost in the number of observations.

Our formulation applies these sequential analysis ideas to a new domain—machine learning inference—where the "time series" is the prediction-increment process of a tree ensemble and the "change-point" signals convergence. Unlike classical settings where the goal is to detect an adverse shift, here the change-point is desirable: it marks the onset of prediction stability, and our stopping rule exploits this transition to terminate computation early.

2.4. Robust Estimation Perspective

Robust statistics [13,30] motivates our use of IQR-based scale in sequential stopping. In particular, FORCE [12] shows how streaming quantiles and IQR-scale estimation provide stable behavior under contamination conditions, which we adapt to prediction-increment trajectories in ensemble inference.

3. Preliminaries and Problem Formulation

3.1. The P^2 Streaming Quantile Algorithm

The P^2 algorithm [11] estimates a streaming quantile $\phi \in (0,1)$ using exactly five markers, requiring $O(1)$ memory and $O(1)$ update time per observation.

Definition 1 (P^2 state)**.** *For a target quantile ϕ, the algorithm maintains five marker heights $q_0 < q_1 < q_2 < q_3 < q_4$, five actual marker positions $n_0 < n_1 < n_2 < n_3 < n_4$, and five desired positions $n'_0 < n'_1 < n'_2 < n'_3 < n'_4$.*

After initialization with the first five sorted observations, the desired positions are set to

$$(n'_0,\ n'_1,\ n'_2,\ n'_3,\ n'_4) = \big(1,\ 1+2\phi,\ 1+4\phi,\ 3+2\phi,\ 5\big), \tag{1}$$

and incremented by $(\delta n_0, \ldots, \delta n_4) = (0,\ \phi/2,\ \phi,\ (1+\phi)/2,\ 1)$ per observation.

When a new observation x arrives, the algorithm: (i) identifies the cell k such that $q_{k-1} \leq x < q_k$; (ii) increments n_j for $j \geq k+1$; and (iii) adjusts interior markers ($i = 1, 2, 3$) using a parabolic formula with a linear fallback:

$$q_i^{\text{new}} = q_i + \frac{d}{n_{i+1} - n_{i-1}} \left[\frac{(n_i - n_{i-1} + d)(q_{i+1} - q_i)}{n_{i+1} - n_i} + \frac{(n_{i+1} - n_i - d)(q_i - q_{i-1})}{n_i - n_{i-1}} \right], \quad (2)$$

where $d = \text{sign}(n_i' - n_i)$. If the parabolic update violates marker ordering, the linear fallback

$$q_i^{\text{new}} = q_i + d \cdot \frac{q_{i+d} - q_i}{n_{i+d} - n_i} \quad (3)$$

is used instead. The quantile estimate is always $\hat{q}_\phi = q_2$ (the central marker).

3.2. FORCE: Robust Scale Estimation via the IQR

The FORCE algorithm [12] uses P^2 to compute streaming quantiles and derives robust statistics as follows.

Definition 2 (Robust scale [12] (Equation (7))). *The IQR-based robust scale estimator for a data stream is*

$$\hat{\sigma} = \frac{\hat{q}_{0.75} - \hat{q}_{0.25}}{1.349}, \quad (4)$$

where $1.349 = 2\,\Phi^{-1}(0.75)$ *ensures Gaussian consistency.*

A key result from FORCE is that IQR-based robust scale estimation inherits a breakdown point of $\varepsilon^* = 0.25$: up to 25% contamination does not invalidate the scale statistic in the Hampel sense [12] (Theorem 1).

Two properties make IQR-based robust scale a natural fit for per-tree increments. First, the 1.349 normalization calibrates the IQR to match the standard deviation for Gaussian data, so FORCE inherits the familiar σ interpretation while dispensing with the Gaussian tail assumption. Second, the 25% breakdown point of the IQR functional means that a moderate fraction of anomalous trees—due to pathological splits, distribution shift in a data shard, or random-output corruption—cannot drive the scale estimate to infinity, providing the detector with an inherent corruption buffer.

3.3. Tree Ensembles and Sequential Prediction

Let $\mathcal{F} = \{f_1, f_2, \ldots, f_T\}$ be a tree ensemble with T base learners. In practice, GBMs are most commonly deployed via modern implementations such as XGBoost [31].

Definition 3 (Ensemble prediction trajectory). *For instance* $\mathbf{x}$*, let* C *denote the number of classes, and let* $\Delta^{C-1} = \{p \in \mathbb{R}^C : p_c \geq 0,\ \sum_c p_c = 1\}$ *denote the* $(C-1)$*-dimensional probability simplex. The cumulative prediction after* t *trees is:*

- *Random Forest (classification):* $\hat{y}_t(\mathbf{x}) = \frac{1}{t}\sum_{s=1}^{t} f_s(\mathbf{x}) \in \Delta^{C-1}$*, where the* c*-th component* $\hat{y}_t^{(c)}(\mathbf{x})$ *is the averaged class-*c *probability;*
- *GBM:* $\hat{y}_t(\mathbf{x}) = \sum_{s=1}^{t} \eta \cdot f_s(\mathbf{x})$*, where* η *is the learning rate.*

Definition 4 (Prediction increment process).

$$\Delta_t(\mathbf{x}) = \hat{y}_t(\mathbf{x}) - \hat{y}_{t-1}(\mathbf{x}), \qquad t \geq 2, \quad \Delta_1 = 0. \quad (5)$$

For RF classification, each tree f_t *outputs a* C*-dimensional vote vector; the cumulative prediction* $\hat{y}_t^{(c)}(\mathbf{x})$ *is the class-*c *probability after averaging* t *trees. We reduce this to a scalar by defining*

$s_t = \max_c \hat{y}_t^{(c)}(\mathbf{x})$ *(the maximum class probability at step t) and* $\Delta_t = s_t - s_{t-1}$ *with* $s_0 = s_1$ *(so* $\Delta_1 = 0$*).*

3.4. Summary of Notation

Table 1 presents the principal symbols used throughout the paper.

Table 1. Summary of notation.

Symbol	Description
T	Total number of trees in the ensemble
t	Tree index ($1 \le t \le T$)
C	Number of classes in the prediction problem
$\mathbf{x}$	Test instance
$\hat{y}_t(\mathbf{x})$	Cumulative ensemble prediction after t trees
$\Delta_t(\mathbf{x})$	Prediction increment at tree t
$\hat{q}_\phi(t)$	P^2 quantile estimate at level ϕ after t observations
$\hat{\sigma}_\Delta(t)$	Streaming IQR scale of increments
W_{ref}	Reference window of tree indices used to estimate $\hat{\sigma}_{\text{ref}}$
$\hat{\sigma}_{\text{ref}}$	Reference scale estimated over W_{ref}
$R(t)$	Relative scale statistic $\hat{\sigma}_\Delta(t)/\hat{\sigma}_{\text{ref}}$
w	Rolling-window size for windowed IQR scale
θ	P^2-STOP convergence threshold
$\hat{\tau}$	Estimated stopping time
τ^*	True change-point
ε	Allowed accuracy degradation in selection constraints

4. Proposed Methods

4.1. P^2-STOP: Nonparametric Ensemble Stopping via Streaming Quantile Change-Point Detection

4.1.1. Convergence Structure of Ensemble Trajectories

We model the prediction-increment process $\{\Delta_t(\mathbf{x})\}_{t=1}^T$ as undergoing a phase transition:

- Phase I (Exploration, $t \le \tau^*$): $\text{Var}(\Delta_t) \ge \sigma^2_{\text{high}}$. Early trees contribute diverse, high-variance predictions.
- Phase II (Convergence, $t > \tau^*$): $\text{Var}(\Delta_t) \le \sigma^2_{\text{low}} \ll \sigma^2_{\text{high}}$. Additional trees provide near-redundant information.

The change-point τ^* marks the transition from exploration to convergence. Detecting τ^* online enables early stopping: once the convergence phase is entered, further trees contribute negligible changes and can be skipped.

4.1.2. Streaming IQR Scale of Increments

Using P^2 estimators for quantiles $\phi = 0.25$ and $\phi = 0.75$ of $\{\Delta_1, \ldots, \Delta_t\}$, we define

Definition 5 (Streaming increment scale)**.**

$$\hat{\sigma}_\Delta(t) = \frac{\hat{q}^{(\Delta)}_{0.75}(t) - \hat{q}^{(\Delta)}_{0.25}(t)}{1.349}, \tag{6}$$

where $\hat{q}^{(\Delta)}_\phi(t)$ *is the* P^2 *estimate of the* ϕ*-quantile of* $\{\Delta_1, \ldots, \Delta_t\}$*.*

4.1.3. Reference Scale and Relative Scale Statistic

Definition 6 (Reference window and reference scale)**.** *Let* $W_{\text{ref}} = \{t_{\text{ref},0}, \ldots, t_{\text{ref},1} - 1\}$ *denote an early reference window (a half-open interval of tree indices) of size* $|W_{\text{ref}}| = t_{\text{ref},1} - t_{\text{ref},0}$*. The*

reference window is fixed before inference and does not adapt online. We estimate the reference scale by averaging scale estimates within the window:

$$\hat{\sigma}_{\text{ref}} = \frac{1}{|W_{\text{ref}}|} \sum_{t \in W_{\text{ref}}} \hat{\sigma}_{\Delta}(t). \tag{7}$$

In our RF experiments with $T = 200$*, we use* $W_{\text{ref}} = [10, 30)$*.*

Definition 7 (Relative scale statistic)**.**

$$R(t) = \frac{\hat{\sigma}_{\Delta}(t)}{\hat{\sigma}_{\text{ref}}}. \tag{8}$$

The stopping rule fires at the first $t \geq t_{\text{begin}}$ such that $R(t) < \theta$, where $t_{\text{begin}} = \max(t_{\min}, t_{\text{ref},1})$ ensures the reference window has completed.

Remark 1 (Heuristic motivation for the threshold range)**.** *While the exact* θ *is selected empirically via validation, the following heuristic motivates the target range (e.g.,* $0.05 \leq \theta \leq 0.10$*).*

RF averaging. For an RF ensemble average of independent or weakly correlated base learners, the variance of the cumulative prediction $\hat{y}_t$ *decays as* $O(1/t)$*. Consequently, the increment* $\Delta_t \approx \frac{1}{t}(f_t - \hat{y}_{t-1})$ *has a standard deviation—and thus an expected IQR scale—that decays proportionally to* $O(1/t)$*. If the reference window* W_{ref} *is centered around early trees (e.g.,* $t_{\text{ref}} \approx 20$*), and the ensemble maximum is* $T = 200$*, the asymptotic scale reduction ratio is expected to be on the order of* $20/200 = 0.10$*.*

GBM. For gradient boosting with learning rate η*, the increment is* $\Delta_t = \eta \cdot f_t(\mathbf{x})$*. The* $O(1/t)$ *decay of increment scale is not generally guaranteed for GBMs: it depends on the residual structure and the base learner capacity. In practice, GBM increments often decrease in magnitude as training loss diminishes, but the rate is dataset- and hyperparameter-specific.*

RF classification experiments. The deployed statistic in our RF experiments is the scalar increment in maximum class probability, $\Delta_t = \max_c \hat{y}_t^{(c)} - \max_c \hat{y}_{t-1}^{(c)}$*, which is a nonlinear transform of the vector-valued cumulative prediction. The* $O(1/t)$ *decay argument applies to the linear average before scalarization and is therefore only an approximate guide for the deployed scalar increment.*

In all settings, the threshold θ *should be selected on a validation fold rather than set purely from the asymptotic argument. The heuristic merely informs the search range.*

4.1.4. The P^2-STOP Algorithm

Complexity

For the full-prefix P^2 variant, P^2-STOP performs a single pass with $O(1)$ work per tree (10 arithmetic operations for each P^2 update) and $O(1)$ memory (10 markers plus a constant number of accumulators). For the default rolling-window variant used in experiments, per-tree cost and memory are $O(w)$ with fixed w (here, $w = 20$), which is effectively constant in this setting. Crucially, to translate these theoretical $O(1)$ or $O(w)$ bounds into wall-clock speedups, P^2-STOP must be deployed in compiled environments (e.g., C++ or Numba). Native Python loops introduce interpreter overhead that negates the benefits of tree reduction, making the Python reference implementation useful only as a logical baseline (see Section 7).

Algorithm 1 presents the complete P^2-STOP procedure.

Algorithm 1 P^2-STOP: nonparametric ensemble stopping

Require: Instance $\mathbf{x}$; ensemble $\mathcal{F} = \{f_1, \ldots, f_T\}$; threshold $\theta \in (0, 1)$; reference window endpoints $t_{\text{ref},0} < t_{\text{ref},1}$; minimum trees $t_{\min}$
Ensure: Stopped prediction $\hat{y}_\tau(\mathbf{x})$; stopping time τ
Initialize P^2 estimators for $\hat{q}_{0.25}$ and $\hat{q}_{0.75}$ of Δ
Initialize accumulators `ref_sum` $\leftarrow 0$, `ref_count` $\leftarrow 0$
for $t = 1, 2, \ldots, T$ **do**
  Compute $\hat{y}_t(\mathbf{x})$ using tree f_t
  if $t = 1$ **then**
    $\Delta_t \leftarrow 0$
  else
    $\Delta_t \leftarrow \hat{y}_t - \hat{y}_{t-1}$
  end if
  Update P^2 estimators with Δ_t
  $\hat{\sigma}_\Delta(t) \leftarrow (\hat{q}_{0.75}(t) - \hat{q}_{0.25}(t))/1.349$
  if $t_{\text{ref},0} \le t < t_{\text{ref},1}$ **then**
    `ref_sum` $\leftarrow$ `ref_sum` $+ \hat{\sigma}_\Delta(t)$
    `ref_count` $\leftarrow$ `ref_count` $+ 1$
  end if
  if $t \ge \max(t_{\min}, t_{\text{ref},1})$ and `ref_count` > 0 **then**
    $\hat{\sigma}_{\text{ref}} \leftarrow$ `ref_sum/ref_count`
    $R(t) \leftarrow \hat{\sigma}_\Delta(t) \,/\, \hat{\sigma}_{\text{ref}}$
    if $R(t) < \theta$ **then**
      return $(\hat{y}_t,\ t)$
    end if
  end if
end for
return $(\hat{y}_T,\ T)$ ▷ Never stopped—use full ensemble

Rolling-Window Scale Variant

The full-prefix P^2 scale (6) accumulates all increments $\{\Delta_1, \ldots, \Delta_t\}$, which includes the high-variance exploration phase and may dilute the signal once convergence begins. An alternative is a rolling-window IQR scale that keeps only the most recent w observations:

$$\hat{\sigma}_\Delta^{(\text{roll})}(t) = \frac{q_{0.75}(W_t) - q_{0.25}(W_t)}{1.349}, \qquad W_t = \{\Delta_{\max(1,t-w+1)}, \ldots, \Delta_t\}, \tag{9}$$

where $q_\phi(W_t)$ denotes the exact empirical ϕ-quantile of the small window ($|W_t| \le w$). Because the window size is fixed (typically $w = 20$), exact percentile computation on the buffer is $O(w)$ per tree—still negligible for moderate w. The rolling-window variant responds more quickly to the exploration-to-convergence transition and is used as the default scale estimator in our experiments; the full-prefix variant is retained as a baseline.

CUSUM-Based Detection

As an alternative to the relative-threshold rule $R(t) < \theta$, we also provide a Page CUSUM detector [26] that monitors the cumulative downward shift:

$$S_t^- = \max\bigl(0,\ S_{t-1}^- + (\mu_{\text{ref}} - \hat{\sigma}_\Delta(t) - k\,\sigma_{\text{ref}})\bigr), \qquad S_0^- = 0, \tag{10}$$

and signals a change-point at the first t where $S_t^- > h\,\sigma_{\text{ref}}$, with allowance k and decision threshold h. The reference statistics μ_{ref} and σ_{ref} are estimated from the scale values in the reference window. This detector adapts automatically to the magnitude of the scale shift, removing the need to manually select θ.

5. Theoretical Analysis

5.1. Heuristic Error Decomposition for the Full-Prefix Variant

The following analysis decomposes the full-prefix scale-estimation error into interpretable components. Because the DKW concentration step treats the nonstationary increment prefix $\{\Delta_1, \ldots, \Delta_t\}$ as if drawn i.i.d. from the mixture distribution $F_{\mathrm{mix},t}$, the resulting bound is a heuristic approximation, not a rigorous finite-sample guarantee. It is retained because it usefully separates three distinct error sources, but the exponential tail bound should be treated as indicative rather than formally valid for the nonstationary case.

Remark 2 (Heuristic error decomposition for the full-prefix variant)**.** *Let $\tilde{\sigma}_\Delta(t)$ denote the IQR scale computed from exact empirical quantiles of the full prefix $\{\Delta_1, \ldots, \Delta_t\}$, and let $\hat{\sigma}_\Delta(t)$ denote the corresponding P^2-based scale. Define:*

- $a_t = |\hat{\sigma}_\Delta(t) - \tilde{\sigma}_\Delta(t)|$ *(P^2 approximation error);*
- $b_t = |\sigma_{\mathrm{mix}}(t) - \sigma_{\mathrm{conv}}|$ *(pre-change contamination bias in the full-prefix scale).*

$\sigma_{\mathrm{mix}}(t)$ is the population IQR scale of the time-t mixture distribution induced by $\{\Delta_1, \ldots, \Delta_t\}$, and σ_{conv} is the convergent-phase population scale.

Assume the mixture density $f_{\mathrm{mix},t}$ is continuous and bounded away from zero in neighborhoods of $q^{\mathrm{mix}}_{0.25}(t)$ and $q^{\mathrm{mix}}_{0.75}(t)$. Then for any $\epsilon > 0$,

$$\mathbb{P}\Big(|\hat{\sigma}_\Delta(t) - \sigma_{\mathrm{conv}}| > \epsilon + a_t + b_t\Big) \le 2\exp\big(-2t\,\epsilon_t'^2\big), \tag{11}$$

where $\epsilon_t' = \epsilon \cdot \min\big(f_{\mathrm{mix},t}(q^{\mathrm{mix}}_{0.25}(t)), f_{\mathrm{mix},t}(q^{\mathrm{mix}}_{0.75}(t))\big)$.

Consequently, for the stopping statistic $R(t) = \hat{\sigma}_\Delta(t)/\hat{\sigma}_{\mathrm{ref}}$, the event $R(t) < \theta$ while the true convergent-scale ratio still exceeds $\theta + (\epsilon + a_t + b_t)/\hat{\sigma}_{\mathrm{ref}}$ is bounded by the same exponential tail.

Heuristic Derivation

We decompose

$$|\hat{\sigma}_\Delta(t) - \sigma_{\mathrm{conv}}| \le |\hat{\sigma}_\Delta(t) - \tilde{\sigma}_\Delta(t)| + |\tilde{\sigma}_\Delta(t) - \sigma_{\mathrm{mix}}(t)| + |\sigma_{\mathrm{mix}}(t) - \sigma_{\mathrm{conv}}| = a_t + \mathcal{E}_t + b_t.$$

The middle term $\mathcal{E}_t$ is controlled by concentration of exact empirical quantiles via DKW [32] and smoothness of the IQR functional; this yields $\mathbb{P}(\mathcal{E}_t > \epsilon) \le 2\exp(-2t\,\epsilon_t'^2)$. Substituting into the decomposition gives (11).

Remark 3. *The decomposition explicitly separates the statistical concentration term (exact empirical quantiles) from the algorithmic approximation term a_t (P^2). Our experiments report empirical agreement between P^2 and exact quantiles to quantify a_t.*

Remark 4 (Heuristic DKW-style bound for the full-prefix variant)**.** *Using the same setup as Remark 2, but without the density assumption on $f_{\mathrm{mix},t}$, a DKW-based bound can be obtained via the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [32], with the tight constant established by Massart [33]. For any continuous distribution F and i.i.d. sample of size t,*

$$\mathbb{P}\big(\sup_x |F_t(x) - F(x)| > u\big) \le 2\exp(-2tu^2). \tag{12}$$

Setting $u = \sqrt{\ln(4/\delta_Q)/(2t)}$, the DKW bound implies that the true quantile q_ϕ lies in the order-statistic interval $\big[X_{(\lceil t(\phi-u)\rceil)},\ X_{(\lfloor t(\phi+u)\rfloor)}\big]$ with probability at least $1 - \delta_Q/2$ for each quantile.

Applying a union bound over the two IQR endpoints ($\phi \in \{0.25, 0.75\}$) yields a data-dependent bound on the IQR-scale error:

$$|\tilde{\sigma}_\Delta(t) - \sigma_{\text{mix}}(t)| \leq \frac{W_{0.75}(t) + W_{0.25}(t)}{1.349} \tag{13}$$

with probability at least $1 - \delta_Q$, where $W_\phi(t) = X_{(\lfloor t(\phi+u) \rfloor)} - X_{(\lceil t(\phi-u) \rceil)}$ is the width of the order-statistic confidence interval for q_ϕ.

Substituting into the decomposition of Remark 2 replaces the density-dependent rate ϵ'_t with the data-adaptive widths $W_\phi(t)$, requiring no knowledge of the underlying density. As with the parent decomposition, this bound inherits the heuristic status of the i.i.d. mixture approximation.

5.2. Formal Stopping Guarantee for the Rolling-Window Variant

Remarks 2 and 4 provide a heuristic decomposition of the full-prefix P^2 scale estimator, where the bound depends on the P^2 approximation error a_t and the pre-change contamination bias b_t—neither of which admits a closed-form expression. Because the full-prefix analysis treats the nonstationary increment sequence as an i.i.d. sample from a mixture distribution, its guarantees are heuristic rather than rigorous. The rolling-window variant (9), used as the default in all experiments, eliminates both nuisance terms by construction and yields a clean finite-sample guarantee.

Theorem 1 (Rolling-window scale concentration)**.** *Consider the rolling-window variant of P^2-STOP with window size w and exact empirical quantiles on the buffer W_t. Suppose $t \geq \tau^* + w$, so that all w observations in W_t are drawn i.i.d. from the convergent-phase increment distribution F_{conv} with continuous density f_{conv}. Let $\sigma_{\text{conv}} = (q^{\text{conv}}_{0.75} - q^{\text{conv}}_{0.25})/1.349$ denote the population convergent-phase scale, and define*

$$f^{\text{conv}}_{\min} = \min\big(f_{\text{conv}}(q^{\text{conv}}_{0.25}),\ f_{\text{conv}}(q^{\text{conv}}_{0.75})\big) > 0.$$

Then for any $\epsilon > 0$,

$$\mathbb{P}\big(|\hat{\sigma}^{(\text{roll})}_\Delta(t) - \sigma_{\text{conv}}| > \epsilon\big) \leq 2\exp\big(-2w\,\epsilon'^2_w\big), \tag{14}$$

where

$$\epsilon'_w = \tfrac{1.349}{2}\, f^{\text{conv}}_{\min} \cdot \epsilon. \tag{15}$$

Proof. Since the rolling-window variant computes exact empirical quantiles on the buffer W_t, the P^2 approximation error vanishes: $a_t = 0$. Since $t \geq \tau^* + w$, the entire window lies in the convergent phase, so the pre-change contamination bias also vanishes: $b_t = 0$. The three-term decomposition of Remark 2 therefore reduces to the single statistical concentration term.

Apply the DKW inequality [32] to the w i.i.d. convergent-phase observations in W_t:

$$\mathbb{P}\big(\sup_x |F_w(x) - F_{\text{conv}}(x)| > u\big) \leq 2\exp(-2wu^2). \tag{16}$$

On the event $\{\sup_x |F_w(x) - F_{\text{conv}}(x)| \leq u\}$, the empirical ϕ-quantile satisfies $|F_{\text{conv}}(\hat{q}_\phi) - \phi| \leq u$. By the mean value theorem applied to F^{-1}_{conv},

$$|\hat{q}_\phi - q^{\text{conv}}_\phi| \leq \frac{u}{f_{\text{conv}}(q^{\text{conv}}_\phi)} \quad \text{for each } \phi \in \{0.25,\ 0.75\}.$$

The IQR error follows by the triangle inequality:

$$|\widehat{\text{IQR}} - \text{IQR}_{\text{conv}}| \leq |\hat{q}_{0.75} - q^{\text{conv}}_{0.75}| + |\hat{q}_{0.25} - q^{\text{conv}}_{0.25}| \leq \frac{2u}{f^{\text{conv}}_{\min}}.$$

Dividing by 1.349,

$$|\hat{\sigma}_{\Delta}^{(\mathrm{roll})}(t) - \sigma_{\mathrm{conv}}| \leq \frac{2u}{1.349 \cdot f_{\mathrm{min}}^{\mathrm{conv}}}.$$

Setting this equal to ϵ and solving for u gives $u = \frac{1.349}{2} f_{\mathrm{min}}^{\mathrm{conv}} \cdot \epsilon = \epsilon'_w$. Substituting into (16) yields (14). □

Corollary 1 (Premature stopping probability for rolling-window variant)**.** *Under the conditions of Theorem 1, suppose the true convergent-phase scale ratio satisfies $\sigma_{\mathrm{conv}}/\hat{\sigma}_{\mathrm{ref}} \geq \theta + \gamma$ for a margin $\gamma > 0$, i.e., the prediction has not yet converged sufficiently relative to the reference. Then the probability that the rolling-window stopping rule falsely fires is bounded by*

$$\mathbb{P}\big(R^{(\mathrm{roll})}(t) < \theta\big) \leq 2\exp\Big(-\tfrac{(1.349)^2}{2}\, w\, \big(f_{\mathrm{min}}^{\mathrm{conv}}\big)^2 \gamma^2\, \hat{\sigma}_{\mathrm{ref}}^2\Big). \tag{17}$$

Proof. If $\sigma_{\mathrm{conv}}/\hat{\sigma}_{\mathrm{ref}} \geq \theta + \gamma$ and $R^{(\mathrm{roll})}(t) = \hat{\sigma}_{\Delta}^{(\mathrm{roll})}(t)/\hat{\sigma}_{\mathrm{ref}} < \theta$, then $|\hat{\sigma}_{\Delta}^{(\mathrm{roll})}(t) - \sigma_{\mathrm{conv}}| > \gamma\,\hat{\sigma}_{\mathrm{ref}}$. Setting $\epsilon = \gamma\,\hat{\sigma}_{\mathrm{ref}}$ in Theorem 1 gives (17). □

Remark 5 (Practical computability)**.** *Bound (17) is fully computable: w, and θ are algorithm parameters, and $\hat{\sigma}_{\mathrm{ref}}$ is computed online from the reference window; γ is a user-specified safety margin; and $f_{\mathrm{min}}^{\mathrm{conv}}$ can be estimated from the buffer W_t via a kernel density estimate at the sample quartiles or conservatively lower-bounded. For the default configuration $w = 20$, $\theta = 0.10$: if $f_{\mathrm{min}}^{\mathrm{conv}} \cdot \gamma \cdot \hat{\sigma}_{\mathrm{ref}} \geq 0.3$, then (17) yields a premature stopping probability below 0.01.*

Remark 6 (Distribution-free alternative)**.** *If the density assumption is undesirable, the DKW-based approach of Remark 4 applies directly to the rolling window. With $a_t = 0$ and $b_t = 0$, the bound on $|\hat{\sigma}_{\Delta}^{(\mathrm{roll})}(t) - \sigma_{\mathrm{conv}}|$ reduces to*

$$\frac{W_{0.75}(w,\delta) + W_{0.25}(w,\delta)}{1.349} \tag{18}$$

with probability at least $1 - \delta$, where $W_{\phi}(w,\delta)$ is the order-statistic confidence-interval width computed from the w buffered observations. This requires no knowledge of f_{conv} and is computable directly from the sorted buffer.

5.3. Asymptotic Discussion

For the full-prefix P^2 variant, one may ask whether the P^2-STOP stopping time is consistent, i.e., whether $\hat{\tau} \xrightarrow{P} \tau^*$ as $T \to \infty$. Using the two-phase model, if (i) $a_t \xrightarrow{P} 0$ (P^2 approximation error vanishes), (ii) $b_t \to 0$ as $t \to \infty$ (contamination bias vanishes), and (iii) $\sigma_{\mathrm{low}}^2/\sigma_{\mathrm{high}}^2 < \theta^2$ (the convergent-phase scale is well below the threshold), then it is plausible that the stopping rule converges to the true change-point. However, a formal proof of this statement would require the establishment of the convergence of the P^2 estimator, which appears to be absent from the literature (see Remark 7 below). We therefore state this as an open conjecture rather than a proved proposition.

For the rolling-window exact variant, conditions (i) and (ii) are automatically satisfied ($a_t = 0$, $b_t = 0$ for $t \geq \tau^* + w$), and the finite-sample guarantee of Theorem 1 provides a direct bound on premature stopping.

Remark 7 (On the P^2 approximation condition)**.** *Condition (i), $a_t \xrightarrow{P} 0$, is empirically well-supported: for continuous distributions the P^2 algorithm has been observed to converge rapidly (within a few hundred observations) in both the original simulations of Jain and Chlamtac [11] and in our own experiments (Section 7). However, a formal proof of L^p or almost-sure convergence of the P^2 estimator appears to be absent from the literature. Our experiments therefore report the empirical*

discrepancy $|a_t| = |\hat{\sigma}_\Delta(t) - \tilde{\sigma}_\Delta(t)|$ *between the* P^2 *and exact quantile-based scales (Section 7.1.3), showing small typical discrepancies with occasional larger trajectory-level peaks.*

5.4. Robustness of the IQR-Scale Functional

Remark 8 (Breakdown point of the IQR-scale functional)**.** *The IQR-based scale functional* $\hat{\sigma} = (\hat{q}_{0.75} - \hat{q}_{0.25})/1.349$ *has breakdown point* $\varepsilon^* = 0.25$ *in the Hampel sense [13]: if at most 25% of observations are adversarially corrupted, the IQR and hence the robust scale estimate remain bounded.*

The full P^2*-STOP stopping rule additionally depends on the reference-window scale estimate* $\hat{\sigma}_{\text{ref}}$*, the ratio statistic* $R(t)$*, and the sequential stopping decision. The exact breakdown point of this composite sequential rule is not formally established in this work. Our contamination experiments (Section 7.3) provide empirical evidence that the stopping rule remains stable with the tested random-tree corruption model (up to 25% contamination), but this empirical stability should not be interpreted as a formal 25% breakdown-point theorem for the overall stopping procedure.*

5.5. Complexity Comparison

Table 2 compares P^2-STOP with the RF Dirichlet baseline.

Table 2. Theoretical properties comparison of ensemble stopping methods.

Method	Assumption	Memory	RF	GBM	Breakdown
Dirichlet–multinomial	Exchangeable categorical votes	$O(C)$	✓	—	N/A
P^2-STOP (ours)	None (nonparametric)	$O(1)$ †	✓	✓	IQR functional: 25% ‡

† $O(1)$ for full-prefix P^2 scale; $O(w)$ for rolling-window variant with window size w. ‡ Breakdown point of the IQR-scale functional; the composite stopping rule's breakdown point is not formally established (see Remark 8).

The measured wall-clock latency in the Python reference implementation, where online P^2-STOP logic is executed in Python loops, shows that tree-count reduction does not consistently translate into latency speedup (Section 7). The Numba-compiled proof-of-concept (Section 7.4.2) shows that wall-clock speedup is attainable once both traversal and stopping logic are executed in a compiled path.

6. Experimental Setup

6.1. Datasets

We evaluated on four publicly available benchmark datasets spanning binary and multiclass classification problems with varying scales. Table 3 summarizes their characteristics.

Table 3. Dataset characteristics.

Dataset	N	p	C	Task	Source
MNIST	70,000	784	10	Multiclass	[34]
Covertype	581,012	54	7	Multiclass	[35]
HIGGS	11,000,000	28	2	Binary	[36]
Credit Card	284,807	30	2	Binary	[37]

For each dataset, we used a 60/10/30 stratified train/validation/test split. The validation fold was used for hyperparameter selection (threshold θ in Phase 1); all reported metrics were computed on the held-out test fold. Up to 20,000 training and 5000 test instances were retained (subsampled where necessary for tractability). Features were standardized to zero mean and unit variance.

6.2. Ensemble Configurations

- Random Forest: `RandomForestClassifier` with $T = 200$ trees, default scikit-learn hyperparameters.

All experiments were repeated over 5 independent random seeds (`random_state` $\in \{42, 123, 456, 789, 1024\}$); see Section 6.7 for details.

6.3. Baselines

1. Full Ensemble: All T trees, no stopping (accuracy ceiling).
2. Dirichlet–Multinomial Stopping (RF only): Bayesian posterior predictive stopping at confidence 0.95.
3. P^2-STOP (proposed): Threshold sweep $\theta \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. Scale estimation uses the rolling-window variant (Equation (9), $w = 20$) with the relative-threshold detection rule by default. Unless otherwise stated, we used warmup $= 10$, $t_{\min} = 10$, and $W_{\text{ref}} = [10, 30)$.

6.4. Adaptive Threshold Selection

Guided by the theoretical bounds discussed in Remark 1, we selected the specific convergence threshold θ automatically from the candidate grid Θ using the validation fold. This formalized the constrained rule already used in our "best per seed" analyses while ensuring the selected threshold remained statistically grounded. Let Θ denote the threshold grid, and let $\Delta\text{Acc}_{\text{val}}(\theta)$ and $\text{WR}_{\text{val}}(\theta)$ denote the validation accuracy change and validation work reduction at threshold θ relative to full evaluation. With tolerance $\varepsilon = 0.005$ (0.5 percentage points), we chose

$$\theta^* \in \arg\max_{\theta \in \Theta} \text{WR}_{\text{val}}(\theta) \quad \text{s.t.} \quad \Delta\text{Acc}_{\text{val}}(\theta) \geq -\varepsilon. \tag{19}$$

Ties were broken by higher validation accuracy, then by smaller θ (more conservative stopping). If no θ satisfied the tolerance, we selected the θ with highest validation accuracy (tie-break by higher work reduction).

6.5. Evaluation Metrics

- Accuracy of the stopped prediction.
- Relative Work $= \bar{\tau}/T$ (mean stopping time normalized by total trees; lower is better).
- Work Reduction $= 1 -$ Relative Work.
- Elbow Fraction: fraction of instances where P^2-STOP triggers before the final tree.
- Pearson/Spearman correlation between P^2-STOP and Dirichlet stopping times.

6.6. Computing Environment

All experiments were executed on a single workstation equipped with an Intel® Xeon® w5-2465X processor (Intel Corporation, Santa Clara, CA, USA) and 128 GB RAM, running Python 3.12 with NumPy 2.3.5, scikit-learn 1.8.0, SciPy 1.16.3, pandas 2.3.3, Numba 0.63.1, and Matplotlib 3.10.8.

6.7. Reproducibility

All experiments were repeated over 5 independent random seeds (`random_state` $\in \{42, 123, 456, 789, 1024\}$), and all uncertainty summaries were computed across these 5 seed-level replicates. For summaries not constrained to $[0, 1]$, we report two-sided 95% Student-t confidence intervals for the mean across seeds; this applies to accuracy, latency, speedup, mean trees used, elbow fraction, correlation summaries, and paired mean differences such as ΔAcc and ΔWR. For summaries constrained to $[0, 1]$, we report

BCa bootstrap 95% confidence intervals for the mean using 20,000 resamples, bootstrap seed 12,345, and clipping to the logical bounds; this applies to work reduction and other bounded-rate summaries. The pairing unit is the seed: for each dataset and configuration, paired differences were computed between methods using the same seed. Reported manuscript p-values are based on two-sided paired t-tests on matched seed-level values, while Wilcoxon signed-rank tests were retained only in the exported CSV files as robustness checks. Accordingly, all result tables in Section 7 use Student-t intervals for accuracy-, difference-, and latency-type summaries and BCa intervals for work-reduction summaries. The P^2 approximation discrepancy analysis (Section 7.1.3) is reported separately as a seed-level mean-discrepancy summary with a 95% confidence interval for mean $|a_t|$. Because only 5 seeds are available, seed-level tests have limited power, and the corresponding p-values should be interpreted cautiously. The threshold-sweep p-values are therefore reported as exploratory and unadjusted for multiple comparisons, with primary emphasis placed on effect sizes and confidence intervals.

7. Results

7.1. Phase 1: Change-Point Detection (P^2-STOP)

7.1.1. Diagnostic Analysis: Existence of the Change-Point

Figure 1 plots the streaming IQR scale $\hat{\sigma}_\Delta(t)$ as a function of tree index for 20 randomly selected instances (gray) with the population mean $\pm$ one standard deviation band (blue), overlaid with the reference threshold $\theta = 0.10$.

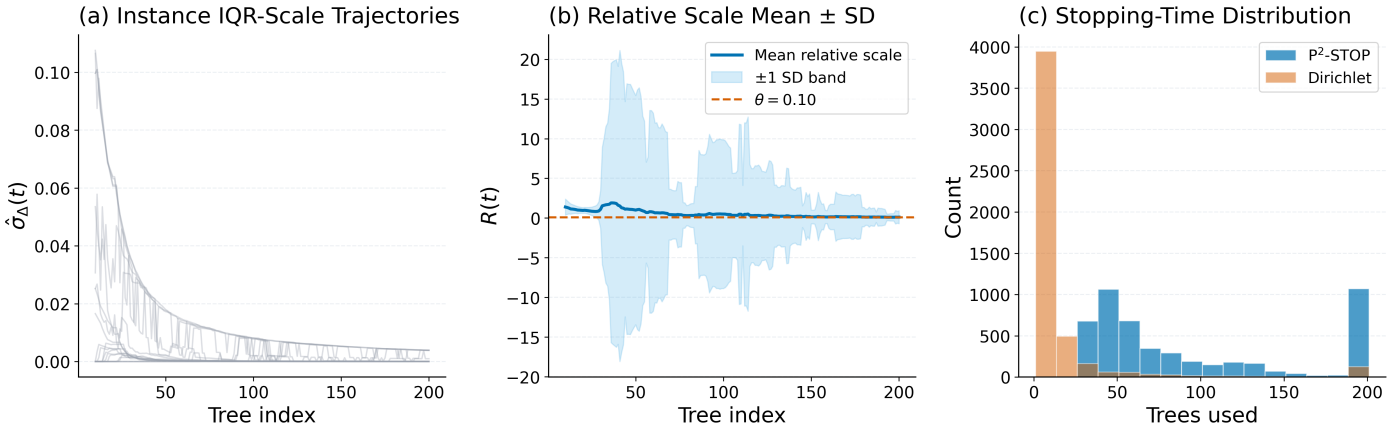


Figure 1. Streaming IQR-scale diagnostics for MNIST with $T = 200$ trees. (**a**) Individual $\hat{\sigma}_\Delta(t)$ curves for 20 instances. (**b**) Population mean $\pm$ standard deviation of the relative scale $R(t)$ with the $\theta = 0.10$ threshold. (**c**) Histogram of stopping times for P^2-STOP vs. Dirichlet baseline.

7.1.2. Threshold Sweep Results

Table 4 reports threshold-sweep outcomes for all four datasets with 95% confidence intervals and seed-level paired tests. These sweep p-values are unadjusted and are reported only as exploratory screening diagnostics across thresholds; no multiple-testing correction was applied because the sweep was not used for confirmatory inference or final threshold selection, and practical interpretation should rely primarily on effect sizes and 95% confidence intervals. This multiplicity caution is specific to the threshold sweeps; the best-per seed, timing, CUSUM, robustness, GBM, and QWYC tables below are matched-seed method comparisons rather than broad threshold-screening families.

Interpreting non-significant results. Several Covertype and HIGGS rows show $p > 0.05$ for the accuracy degradation column. With only five seeds, a non-significant paired test means the difference was not detected at the given sample size—not that no difference exists. The 95% confidence intervals, which span both negative and mildly

positive values, are more informative than the p-values alone: they bound the plausible effect size and should be used for practical decision-making.

Table 4. P^2-STOP threshold sweep (MNIST, Covertype, HIGGS, and Credit Card) with confidence intervals and paired tests (5 seeds).

Dataset	θ	ΔAcc vs. Full [95% CI]	p (Paired)	WR(P^2) [95% CI]	p for ΔWR(P^2 − Dir.)
MNIST	0.05	−0.0040 [−0.0056, −0.0025]	1.82×10^{-3}	0.450 [0.446, 0.455]	1.89×10^{-9}
	0.10	−0.0050 [−0.0059, −0.0041]	9.35×10^{-5}	0.530 [0.524, 0.536]	9.87×10^{-9}
	0.15	−0.0061 [−0.0065, −0.0056]	3.05×10^{-6}	0.573 [0.566, 0.580]	2.65×10^{-8}
	0.20	−0.0071 [−0.0078, −0.0063]	1.17×10^{-5}	0.599 [0.592, 0.606]	5.21×10^{-8}
	0.25	−0.0071 [−0.0089, −0.0052]	4.57×10^{-4}	0.616 [0.609, 0.623]	6.32×10^{-8}
	0.30	−0.0073 [−0.0089, −0.0057]	2.40×10^{-4}	0.628 [0.621, 0.635]	7.93×10^{-8}
Covertype	0.05	−0.0007 [−0.0014, −0.0000]	4.49×10^{-2}	0.516 [0.511, 0.522]	7.01×10^{-11}
	0.10	−0.0019 [−0.0036, −0.0002]	3.44×10^{-2}	0.612 [0.607, 0.617]	1.61×10^{-10}
	0.15	−0.0031 [−0.0064, +0.0001]	5.52×10^{-2}	0.664 [0.659, 0.669]	7.16×10^{-11}
	0.20	−0.0041 [−0.0084, +0.0001]	5.39×10^{-2}	0.695 [0.691, 0.699]	3.38×10^{-10}
	0.25	−0.0053 [−0.0107, +0.0000]	5.10×10^{-2}	0.716 [0.713, 0.720]	1.57×10^{-10}
	0.30	−0.0054 [−0.0080, −0.0028]	4.74×10^{-3}	0.731 [0.727, 0.734]	4.73×10^{-11}
HIGGS	0.05	−0.0006 [−0.0022, +0.0009]	3.14×10^{-1}	0.467 [0.456, 0.479]	3.89×10^{-8}
	0.10	−0.0022 [−0.0042, −0.0002]	3.84×10^{-2}	0.588 [0.580, 0.597]	2.98×10^{-7}
	0.15	−0.0023 [−0.0065, +0.0018]	1.97×10^{-1}	0.663 [0.656, 0.669]	5.99×10^{-6}
	0.20	−0.0054 [−0.0107, −0.0001]	4.87×10^{-2}	0.708 [0.703, 0.712]	1.67×10^{-4}
	0.25	−0.0070 [−0.0140, +0.0000]	5.07×10^{-2}	0.738 [0.735, 0.742]	1.18×10^{-2}
	0.30	−0.0073 [−0.0123, −0.0024]	1.49×10^{-2}	0.760 [0.757, 0.762]	3.91×10^{-1}
Credit Card	0.05	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.005 [0.004, 0.006]	9.61×10^{-13}
	0.10	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.006 [0.004, 0.007]	1.59×10^{-12}
	0.15	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.006 [0.005, 0.007]	1.87×10^{-12}
	0.20	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.006 [0.005, 0.008]	1.93×10^{-12}
	0.25	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.007 [0.005, 0.008]	1.99×10^{-12}
	0.30	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	0.007 [0.005, 0.008]	2.12×10^{-12}

On the highly imbalanced Credit Card dataset (Table 4, bottom block), predictions stabilized almost immediately: ΔAcc is statistically indistinguishable from zero ($p = 0.37$) at every threshold, and work reduction remains below 1%. P^2-STOP correctly identifies near-zero improvability and effectively uses the Full Ensemble.

7.1.3. P^2 Approximation Error vs. Exact Quantiles

To close the theory-to-experiment loop for the asymptotic discussion's condition (i) (see Remark 7), $a_t \xrightarrow{P} 0$, we computed $|a_t| = |\hat{\sigma}_\Delta(t) - \tilde{\sigma}_\Delta(t)|$ directly from saved Phase 1 increment streams. For each dataset and seed, we subsampled 250 test instances and compared the full-prefix P^2 IQR scale with the full-prefix exact empirical IQR scale at each tree index. Table 5 reports the results.

Across datasets, average discrepancies are small (largest mean $|a_t| = 0.0068$ on HIGGS), supporting the empirical plausibility of condition (i). Rare larger deviations appear in worst-case trajectories ($\max |a_t| \approx 0.091$), but they are not representative of the typical per-stream error profile.

Table 5. Empirical discrepancy between P^2 and exact full-prefix quantile scales (a_t) (5 seeds, 250 instances/seed/dataset).

Dataset	Max $\lvert a_t \rvert$	Mean $\lvert a_t \rvert$ [95% CI]	Mean Per-Stream $q_{0.95}(\lvert a_t \rvert)$
MNIST	0.0889	0.0039 [0.0036, 0.0042]	0.0087
Covertype	0.0910	0.0054 [0.0048, 0.0060]	0.0110
HIGGS	0.0910	0.0068 [0.0064, 0.0072]	0.0139
Credit Card	0.0558	2.8×10^{-5} [1.7×10^{-5}, 3.9×10^{-5}]	9.4×10^{-5}

End-to-End Impact on Stopping

Although the full-prefix P^2 estimator tracks the exact IQR closely, it operates on the entire history, so the scale estimate is dominated by exploration-phase variability and decays slowly. The rolling-window variant, by contrast, conditions only on the most recent w increments and therefore responds quickly to convergence. Table 6 compares the two backends under identical conditions ($\theta = 0.10$, five seeds).

Table 6. Comparison of full-prefix P^2 backend vs. rolling-window exact backend ($\theta = 0.10$, five seeds). Both backends use the same stopping rule; only the scale estimation differs.

Dataset/Model	Backend	ΔAcc vs. Full	Work Reduction	WR [95% CI]
MNIST/RF	Full-prefix P^2	−0.0012	0.123	[0.119, 0.125]
MNIST/RF	Rolling-window	−0.0050	0.530	[0.526, 0.533]
Covertype/RF	Full-prefix P^2	+0.0000	0.130	[0.128, 0.132]
Covertype/RF	Rolling-window	−0.0019	0.612	[0.610, 0.617]
MNIST/XGBoost	Full-prefix P^2	−0.0009	0.323	[0.301, 0.347]
MNIST/XGBoost	Rolling-window	−0.0033	0.670	[0.665, 0.675]

The rolling-window backend consistently achieves 2–5× higher work reduction than the full-prefix P^2 backend because its local scale estimate tracks the convergence transition more responsively. The accuracy cost is modest (<0.5 pp additional on MNIST/RF). This confirms our recommendation of the rolling-window variant as the practical default.

7.1.4. Comparison with Parametric Baseline (Dirichlet)

Table 7 reports best per seed P^2-STOP versus Dirichlet with CIs and paired tests. For each dataset/seed, the per seed threshold θ^* is selected on the validation fold by the automatic rule in Section 6.4; the table reports held-out test performance at the selected thresholds.

Table 7. Best per seed P^2-STOP versus Dirichlet baseline (5 seeds).

Dataset	θ^* Mode	ΔAcc(P^2 − Dir.) [95% CI]	p (Paired)	ΔWR(P^2 − Dir.) [95% CI]	p (Paired)	Pearson r [95% CI]
MNIST	0.05	−0.0048 [−0.0065, −0.0030]	1.53×10^{-3}	−0.399 [−0.491, −0.307]	2.77×10^{-4}	−0.109 [−0.144, −0.075]
Covertype	0.30	−0.0020 [−0.0048, +0.0007]	1.08×10^{-1}	−0.151 [−0.189, −0.114]	3.53×10^{-4}	−0.022 [−0.102, +0.058]
HIGGS	0.05	+0.0032 [−0.0008, +0.0072]	9.30×10^{-2}	−0.188 [−0.311, −0.064]	1.37×10^{-2}	+0.371 [+0.273, +0.468]
Credit Card	0.30	+0.0000 [−0.0001, +0.0002]	3.74×10^{-1}	−0.943 [−0.945, −0.941]	2.12×10^{-12}	−0.240 [−0.307, −0.173]

At best per seed configurations, mean work reduction versus full RF is 44.8% across datasets (range: 0.7–71.7%), with an accuracy change in the range −0.53 to +0.02 percentage

points. Against Dirichlet, P^2-STOP yields lower work reduction on all four datasets in this run, consistent with our positioning that the primary contribution is nonparametric robustness and model-agnosticity rather than RF-specific work-reduction dominance.

7.1.5. Accuracy–Work Pareto Frontier (All Datasets)

Figure 2 shows the accuracy vs. work-reduction trade-off across all thresholds and datasets.

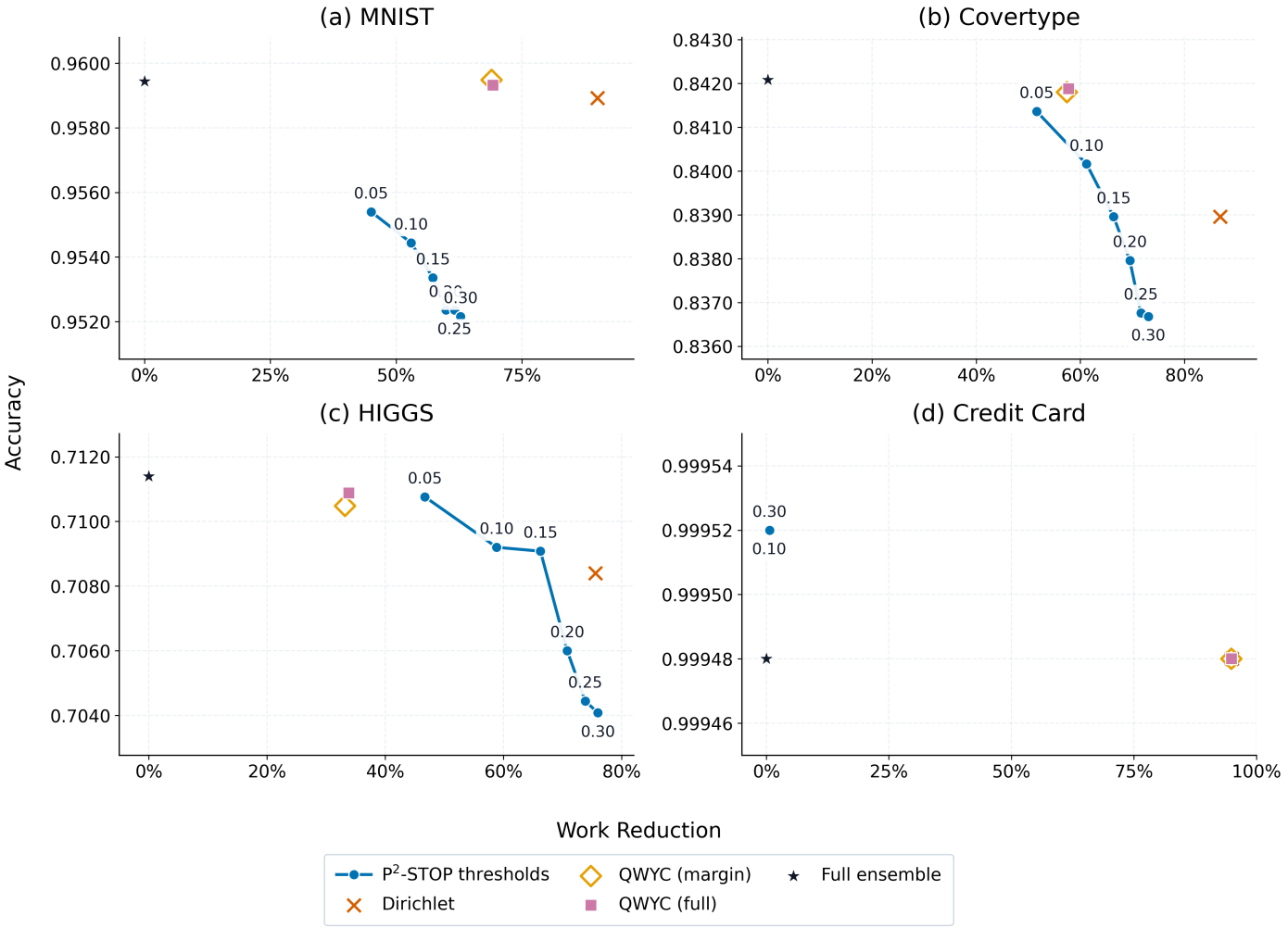


Figure 2. 2 × 2 Pareto frontiers across MNIST, Covertype, HIGGS, and Credit Card. P^2-STOP thresholds are shown as the blue curve with point labels $\theta \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$; Dirichlet is marked by ×; QWYC margin-only [6] by diamonds; QWYC-full with optimized ordering by squares; and the Full Ensemble by a black star.

7.1.6. Comparison with QWYC Baseline (Margin-Only and Full)

Wang et al. [6] propose QWYC ("Quit When You Can"), which combines (i) an optimized evaluation ordering and (ii) a margin-based early-exit criterion: stop evaluating when the running margin exceeds the maximum change remaining according to the model's update rule. For RF the bound is $m_t > (T - t)/T$ where m_t is the margin at tree t; for GBM, the analogous bound incorporates the learning rate.

We evaluated QWYC in two configurations:

- QWYC-margin: Component (ii) alone, with the natural tree order—this isolates the stopping-rule comparison.
- QWYC-full: Components (i) + (ii)—the greedy forward-selection ordering of Wang et al. [6], specialized to equal-weight RF: at each position, the remaining tree is

selected maximizing the count of margin-stopped validation instances, followed by margin stopping on the reordered trajectory. Ordering optimization applies only to RF (independently trained trees are freely permutable); for XGBoost, trees are sequential residual correctors and cannot be reordered, so QWYC-full ≡ QWYC-margin.

Table 8 reports matched-seed accuracy and work-reduction comparisons ($\theta = 0.10$ for P^2-STOP, 5 seeds).

Table 8. Comparison of P^2-STOP ($\theta = 0.10$) against QWYC [6] in two configurations: QWYC-margin (component (ii), fixed tree order) and QWYC-full (components (i) + (ii), validation-optimized ordering plus margin stop). For XGBoost, ordering optimization is not applicable, so QWYC-full ≡ QWYC-margin. (5 seeds, seed-level paired tests).

Dataset	Model	Acc (Full)	Acc (QWYC-m)	Acc (QWYC-f)	Acc (P^2)	WR (QWYC-m)	WR (QWYC-f)	WR (P^2)	ΔWR (P^2 − Qf)	p (Paired)
MNIST	RF	0.9594	0.9595	0.9593	0.9544	0.690	0.693	0.530	−0.163	7.95×10^{-7}
MNIST	XGBoost	0.9543	0.9543	0.9543	0.9510	0.801	0.801	0.670	−0.131	9.16×10^{-8}
Covertype	RF	0.8421	0.8418	0.8419	0.8402	0.574	0.578	0.612	+0.034	7.74×10^{-5}
Covertype	XGBoost	0.8059	0.8048	0.8048	0.7884	0.541	0.541	0.554	+0.013	1.54×10^{-1}
HIGGS	RF	0.7114	0.7105	0.7109	0.7092	0.332	0.338	0.588	+0.250	5.05×10^{-8}
HIGGS	XGBoost	0.6917	0.6918	0.6918	0.6982	0.437	0.437	0.584	+0.147	2.90×10^{-4}
Credit Card	RF	0.9995	0.9995	0.9995	0.9995	0.949	0.949	0.006	−0.943	2.32×10^{-12}
Credit Card	XGBoost	0.9995	0.9995	0.9995	0.9995	0.942	0.942	0.857	−0.085	1.07×10^{-9}

The comparison reveals complementary strengths. QWYC-full, which combines validation-optimized tree ordering with exact model-specific margin bounds, achieves the highest work reduction on datasets where high margins trigger early stopping (e.g., MNIST and Credit Card). The ordering optimization provides a measurable uplift over QWYC-margin on RF datasets. P^2-STOP achieves higher work reduction on HIGGS (both models) and Covertype RF, where gradual convergence favors the data-driven scale approach over QWYC's conservative margin bound. Crucially, P^2-STOP requires no model-specific margin derivation and no ordering-optimization pass, making it immediately applicable to arbitrary ensembles out of the box, whereas QWYC's margin bound must be re-derived for each model family and its ordering optimizer requires a dedicated validation pass. The updated Pareto frontiers in Figure 2 visualize these complementary trade-offs.

7.2. CUSUM Variant Evaluation

Equation (10) introduces a Page-style CUSUM detector as an alternative to the default relative-threshold rule. To close the method-to-results loop, we evaluated CUSUM directly against the relative rule using identical seeds and data splits (5 seeds; datasets: MNIST/Covertype/HIGGS/Credit Card). CUSUM uses fixed parameters $k = 0.5$ and $h = 4.0$ with the same rolling scale settings ($w = 20$, warmup $= 10$), minimum-trees constraint ($t_{\min} = 10$), and reference window ($W_{\text{ref}} = [10, 30)$). The relative baseline uses the best per seed configuration from Table 7. Table 9 reports the paired seed-level comparison.

Table 9. CUSUM detector versus relative-threshold detector (paired seed-level comparison, 5 seeds). Positive ΔWR means CUSUM uses fewer trees.

Dataset	ΔAcc (CUSUM − Rel.) [95% CI]	p	ΔWR (CUSUM − Rel.) [95% CI]	p
MNIST	−0.0016 [−0.0034, +0.0001]	6.16×10^{-2}	+0.1167 [+0.0193, +0.2140]	2.92×10^{-2}
Covertype	−0.0020 [−0.0053, +0.0013]	1.72×10^{-1}	+0.0066 [−0.0310, +0.0442]	6.51×10^{-1}
HIGGS	−0.0127 [−0.0169, −0.0085]	1.10×10^{-3}	+0.2232 [+0.0969, +0.3495]	8.01×10^{-3}
Credit Card	−0.0000 [−0.0002, +0.0001]	3.74×10^{-1}	−0.0010 [−0.0011, −0.0009]	1.79×10^{-5}

The CUSUM variant is therefore not uniformly superior. It provides a clear work-reduction gain on HIGGS (with measurable accuracy drop) and also increases work reduction on MNIST with only a small, statistically weak accuracy decrease. Differences are statistically weak on Covertype and slightly unfavorable on Credit Card.

7.3. Contamination Robustness Stress Test

To empirically probe the IQR breakdown-point property discussed in Remark 8, we ran a contamination stress test in which 5%, 15%, and 25% of trees were replaced by random-output trees (Dirichlet draws over class probabilities). We compared the default IQR-based detector (P^2-STOP) with a non-robust standard-deviation (SD) scale baseline with identical seeds and data splits.

Using this random-tree corruption model, P^2-STOP does not exhibit catastrophic degradation up to 25% contamination: across datasets and contamination levels, its accuracy relative to the corresponding contaminated Full Ensemble remains within 0.55 percentage points. However, the paired IQR-versus-SD differences are mostly not statistically significant with five seeds. Empirically, this supports stability under contamination conditions more clearly than uniform dominance over the SD-scale baseline. Table 10 reports the paired seed-level results.

Table 10. Contamination stress test: paired seed-level comparison between IQR-scale and SD-scale detectors (5 seeds). Positive deltas favor IQR.

Dataset	Corrupt %	ΔAcc (IQR−SD) [95% CI]	*p*	ΔWR (IQR−SD) [95% CI]	*p*
MNIST	5	−0.0010 [−0.0044, +0.0025]	4.86×10^{-1}	−0.0623 [−0.3045, +0.1800]	5.15×10^{-1}
	15	−0.0002 [−0.0036, +0.0032]	8.78×10^{-1}	+0.0707 [−0.2732, +0.4146]	5.99×10^{-1}
	25	+0.0016 [−0.0016, +0.0047]	2.38×10^{-1}	−0.0382 [−0.3047, +0.2283]	7.11×10^{-1}
Covertype	5	+0.0001 [−0.0038, +0.0040]	9.36×10^{-1}	+0.0612 [−0.0655, +0.1879]	2.51×10^{-1}
	15	+0.0006 [−0.0016, +0.0028]	4.99×10^{-1}	+0.0683 [−0.0349, +0.1715]	1.40×10^{-1}
	25	−0.0024 [−0.0056, +0.0007]	9.52×10^{-2}	+0.1372 [−0.1364, +0.4108]	2.36×10^{-1}
HIGGS	5	+0.0069 [−0.0078, +0.0215]	2.63×10^{-1}	−0.0345 [−0.4368, +0.3677]	8.23×10^{-1}
	15	+0.0074 [−0.0053, +0.0201]	1.83×10^{-1}	−0.0269 [−0.5164, +0.4627]	8.86×10^{-1}
	25	+0.0081 [−0.0036, +0.0199]	1.28×10^{-1}	−0.0377 [−0.4361, +0.3607]	8.06×10^{-1}
Credit Card	5	+0.0000 [+0.0000, +0.0000]	—	−0.2713 [−0.6702, +0.1275]	1.32×10^{-1}
	15	+0.0001 [−0.0001, +0.0002]	1.78×10^{-1}	−0.0406 [−0.2291, +0.1478]	5.82×10^{-1}
	25	+0.0000 [−0.0002, +0.0002]	1.00	+0.0263 [−0.0015, +0.0541]	5.82×10^{-2}

7.4. End-to-End Inference Timing

7.4.1. Logical Baseline: Python Loop Overhead

Table 11 reports per-instance inference latency on the same prediction path for full and early-stop inference (tree traversal, online stopping logic, and prediction output all included). Statistics were aggregated over five seeds with 95% confidence intervals; paired tests compared each stopping method against full-ensemble latency.

These results reflect latency in the native Python reference implementation. As anticipated, because the Python interpreter adds substantial overhead to the online stopping loop, P^2-STOP does not achieve consistent wall-clock speedup versus full RF in this uncompiled prototype. This establishes the Python environment strictly as a logical baseline for tree-usage reduction, rather than a practical deployment target.

Table 11. Per-instance end-to-end latency and speedup relative to Full Ensemble evaluation (RF; Python reference implementation).

Dataset	Method	ms/Instance (95% CI)	Speedup vs. Full (95% CI)	p vs. Full
MNIST	Full Ensemble (RF)	0.0688 [0.0677, 0.0698]	1.000× [1.000, 1.000]	—
	P^2-STOP	0.0944 [0.0720, 0.1168]	0.752× [0.561, 0.942]	3.43×10^{-2}
	Dirichlet	0.0278 [0.0261, 0.0295]	2.477× [2.365, 2.590]	1.05×10^{-8}
Covertype	Full Ensemble (RF)	0.0458 [0.0416, 0.0500]	1.000× [1.000, 1.000]	—
	P^2-STOP	0.0454 [0.0394, 0.0513]	1.018× [0.858, 1.177]	8.96×10^{-1}
	Dirichlet	0.0268 [0.0249, 0.0287]	1.711× [1.521, 1.900]	3.27×10^{-4}
HIGGS	Full Ensemble (RF)	0.0392 [0.0368, 0.0416]	1.000× [1.000, 1.000]	—
	P^2-STOP	0.0633 [0.0457, 0.0809]	0.647× [0.447, 0.847]	2.34×10^{-2}
	Dirichlet	0.0329 [0.0324, 0.0334]	1.192× [1.111, 1.274]	2.56×10^{-3}
Credit Card	Full Ensemble (RF)	0.0101 [0.0095, 0.0107]	1.000× [1.000, 1.000]	—
	P^2-STOP	0.0702 [0.0695, 0.0709]	0.144× [0.136, 0.152]	2.67×10^{-9}
	Dirichlet	0.0029 [0.0013, 0.0045]	3.849× [2.365, 5.332]	2.54×10^{-4}

7.4.2. Target Deployment: Compiled Inference Speedup (Numba)

To demonstrate P^2-STOP in its intended deployment environment, we evaluated a Numba-compiled implementation where both tree traversal and the online stopping loop execute in a highly optimized compiled path (excluding one-time JIT compilation overhead via warmup runs). To avoid confounding by loop-order effects, the Full-Numba baseline uses the same tree-major traversal ordering as the P^2-STOP kernel; the only difference is whether stopping computations and early termination are applied. For this scalability check we used $T = 500$ trees (same datasets and seeds) and retained the validation-selected thresholds from Section 6.4. Table 12 reports end-to-end per-instance latency aggregated over five seeds.

Table 12. Numba-compiled end-to-end timing (RF, $T = 500$). Speedup is relative to Full Ensemble with the same Numba engine.

Dataset	Full-Numba ms/inst. (95% CI)	P^2-STOP-Numba ms/inst. (95% CI)	Speedup vs. Full-Numba (95% CI)	WR (95% CI)	p vs. Full-Numba
MNIST	0.1530 [0.1513, 0.1547]	0.0366 [0.0320, 0.0413]	4.215× [3.641, 4.790]	0.680 [0.664, 0.701]	2.85×10^{-7}
Covertype	0.1028 [0.1015, 0.1042]	0.0215 [0.0187, 0.0244]	4.820× [4.235, 5.404]	0.851 [0.832, 0.861]	1.20×10^{-7}
HIGGS	0.0764 [0.0730, 0.0798]	0.0197 [0.0124, 0.0269]	4.142× [2.760, 5.524]	0.820 [0.781, 0.858]	7.29×10^{-6}
Credit Card	0.0136 [0.0108, 0.0164]	0.0241 [0.0228, 0.0255]	0.562× [0.475, 0.648]	0.010 [0.007, 0.012]	4.33×10^{-5}

The Numba-compiled results with $T = 500$ clarify that wall-clock speedup is attainable once both tree traversal and the stopping loop are executed with the same compiled engine: P^2-STOP yields consistent speedups on MNIST/Covertype/HIGGS (4.14×–4.82×, Table 12). Conversely, when stopping rarely triggers (Credit Card: work reduction around 1%), the additional stopping computations dominate even in a compiled path, yielding slowdown.

7.5. Cross-Model Validation on GBM (XGBoost)

To validate the model-agnostic claim empirically, we ran P^2-STOP on XGBoost classifiers with $T = 500$ boosting rounds using the same threshold grid $\theta \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ and seed-level selection criterion used in RF. Thresholds were selected on the validation fold by the same automatic rule (Equation (19)). Table 13 reports the results.

Table 13. GBM cross-model validation across four datasets (XGBoost, $T = 500$): best per seed P^2-STOP vs. full GBM (5 seeds).

Dataset	θ^* Mode	ΔAcc (P^2 − Full) [95% CI]	p (Paired)	WR(P^2) [95% CI]
MNIST	0.10	-0.0027 $[-0.0046, -0.0008]$	1.82×10^{-2}	0.691 [0.667, 0.717]
Covertype	0.05	-0.0110 $[-0.0168, -0.0052]$	6.13×10^{-3}	0.414 [0.386, 0.427]
HIGGS	0.30	$+0.0046$ $[-0.0102, +0.0194]$	4.36×10^{-1}	0.816 [0.809, 0.822]
Credit Card	0.30	-0.0001 $[-0.0004, +0.0002]$	3.74×10^{-1}	0.890 [0.886, 0.895]

Across four datasets, the GBM results preserve the core pattern: work reduction is substantial on MNIST/HIGGS/Credit Card, while Covertype exhibits a less favorable trade-off. In fact, on GBM Covertype no threshold in our tested grid simultaneously achieves small accuracy loss and large work reduction; the selected setting ($\theta^* = 0.05$) still incurs a non-negligible accuracy drop with moderate work reduction.

This behavior is also substantively informative: boosted Covertype trajectories appear less cleanly two-phase than MNIST/HIGGS in this setup. A plausible mechanism is persistent late-round margin correction on heterogeneous multiclass boundary regions, where additional trees continue refining hard instances rather than collapsing rapidly to near-zero increments.

7.6. Ablation: Sensitivity to the Reference Window

The reference scale $\hat{\sigma}_{\text{ref}}$ anchors the relative-scale statistic $R(t)$ and is estimated over an early reference window W_{ref} (Equation (7)). To assess sensitivity, we varied W_{ref} while keeping the stopping threshold $\theta = 0.10$ fixed. Table 14 reports the resulting accuracy and work reduction.

Table 14. Sensitivity of P^2-STOP to the reference window W_{ref} ($\theta = 0.10$). Window endpoints are reported as half-open intervals $[t_0, t_1)$.

Dataset	[5, 10)	[10, 20)	[10, 30)	[20, 40)
Accuracy (%)				
MNIST	95.26	95.38	95.44	95.54
Covertype	83.83	83.87	84.02	84.10
HIGGS	70.78	70.84	70.92	71.02
Credit Card	99.95	99.95	99.95	99.95
Work Reduction (%)				
MNIST	45.49	51.15	52.97	49.97
Covertype	54.30	59.75	61.20	56.30
HIGGS	60.26	60.68	58.85	50.24
Credit Card	0.28	0.43	0.58	0.67

Earlier/shorter reference windows use fewer scale observations and can yield a noisier $\hat{\sigma}_{\text{ref}}$, which can shift triggering earlier or later depending on the realized reference level. Later/longer windows stabilize $\hat{\sigma}_{\text{ref}}$ but can incorporate convergent-phase behavior, delaying triggering. The newly tested $[10, 30)$ window achieves the highest work reduction on MNIST (52.97%) and Covertype (61.20%), confirming it as a practical sweet spot that balances reference-scale stability against convergence-phase contamination. HIGGS and Credit Card show monotonically increasing WR with wider windows but diminishing returns; the accuracy differences across all four windows are small ($\leq$0.28 pp on any dataset). The default $W_{\text{ref}} = [10, 30)$ used in the main experiments therefore balances these effects well.

8. Discussion

8.1. Nonparametric and Model-Agnostic Positioning

The empirical results indicate that P^2-STOP provides a nonparametric stopping rule that substantially reduces tree-evaluation work versus full-ensemble inference, with a dataset-dependent but generally modest accuracy change. The rolling-window variant—used as the default in all experiments—admits a clean finite-sample guarantee (Theorem 1 and Corollary 1): because it computes exact quantiles on a fixed-size buffer, both the P^2 approximation error and the pre-change contamination bias vanish by construction, yielding a computable upper bound on premature-stopping probability that decreases exponentially in w. The contribution is methodological generality and robustness, not superior RF-specific work reduction versus Dirichlet stopping. The CUSUM evaluation (Table 9) shows detector-dependent trade-offs rather than universal dominance. The contamination stress test (Table 10) also shows mixed IQR-versus-SD differences; in this setting, the evidence supports stability up to 25% corruption but not a blanket claim that a SD-scale baseline fails across all datasets.

The QWYC comparison (Table 8) further clarifies this positioning. We evaluated QWYC in two configurations: QWYC-margin (stopping criterion only, with the natural tree order) and QWYC-full (validation-optimized ordering plus margin stop). On high-margin datasets (MNIST and Credit Card), QWYC-full achieves the highest work reduction, with the ordering optimization providing a measurable uplift over QWYC-margin alone. On datasets where margins converge gradually (HIGGS, Covertype), the data-driven IQR-scale approach of P^2-STOP dominates both QWYC variants. Crucially, P^2-STOP requires no model-specific margin derivation and no ordering-optimization pass, making it immediately applicable to arbitrary ensemble types out of the box.

Crucially, the translation of algorithmic work reduction to true wall-clock speedup dictates that P^2-STOP be used in compiled inference engines. End-to-end timing in native Python is dominated by interpreter overhead (Table 11), positioning Python purely as a logical baseline. However, when deployed in its intended compiled environment (Numba), P^2-STOP demonstrates substantial and consistent wall-clock speedups (4.14×–4.82× on MNIST/Covertype/HIGGS, Table 12). However, the effect remains dataset-dependent and can reverse when stopping rarely triggers (Credit Card).

8.2. Practitioner's Guide

Table 15 summarizes recommended defaults and practical guidance for applying P^2-STOP.

For latency-sensitive deployments, wall-clock speedup typically requires that the stopping logic be implemented in a compiled inference path (e.g., Numba/C/C++); otherwise, Python loop overhead can dominate even when tree-count work reduction is large (Table 11 vs. Table 12).

Table 15. Practitioner's guide: recommended defaults and when to expect good performance.

Item	Recommended Default	Guidance
Threshold θ (relative-scale rule)	0.10	Based on the $O(1/t)$ heuristic for increment-variance decay (Remark 1), 0.10 is a suggested starting point for $T = 200$ and early reference windows. A validation set with an accuracy tolerance (e.g., $\Delta\text{Acc}_{\text{val}} \geq -0.5$ percentage points) is employed for tuning using the automatic rule in (19). Smaller θ is more conservative (later stopping); larger θ yields more work reduction but can increase accuracy loss.
Rolling window w (Equation (9))	20	Small w reacts quickly but is noisier; large w is smoother but can delay detection. Fixed w keeps per-tree cost effectively constant.
Warmup/minimum samples	10	Require at least five observations for quartiles; using 10 improves stability of the rolling IQR scale early in the trajectory.
Reference window W_{ref} (Equation (7))	$[10, 30)$	Choose an early window that captures exploration variability but precedes convergence. Too-early windows can be unstable; too-late windows can include convergent behavior and weaken the contrast.
Minimum trees $t_{\min}$	10	Enforces a minimum evaluation budget and avoids stopping on transient early fluctuations. We begin detection at $t_{\text{begin}} = \max(t_{\min}, t_{\text{ref},1})$.
Detector variant	Relative-threshold (default)	The Page-style CUSUM detector (Equation (10)) can be useful when the scale decay is gradual and thresholding is sensitive; in our experiments, it increased work reduction on HIGGS at the cost of measurable accuracy drop and was not uniformly superior.
Expected good regimes	—	Clear exploration-to-convergence transition in the rolling IQR scale (a sharp drop in $R(t)$) and a high elbow fraction; these are the regimes where large work reduction is most likely.
Be cautious when	—	Stopping rarely triggers (work reduction ≈ 0), as in highly imbalanced or near-deterministic problems; or when the scale decays gradually with no clear change-point (threshold becomes more sensitive). In such cases, expect limited savings and validate accuracy carefully.

8.3. Connection to Robust Statistics and FORCE

P^2-STOP inherits robustness from IQR-based streaming scale estimation, consistent with the FORCE paradigm [12]. In our contamination experiment, this robustness appears as stable performance with injected random-tree corruption; however, superiority over an SD-scale detector is dataset-dependent.

8.4. Limitations and Future Directions

1. The change-point model assumes a dominant exploration-to-convergence transition; some instances may show weak or multi-stage transitions.
2. Threshold selection remains data-dependent. We mitigate this with validation-based selection and report uncertainty with CIs and paired tests, but fully adaptive thresholding—potentially via Shiryaev-style Bayes-optimal stopping rules [22,24]—remains open.
3. The contamination stress test uses random-output tree replacement; stronger adversarial contamination models are an important next robustness benchmark.
4. The current benchmark centers on classification. Extending the same stopping logic to broader regression benchmarks is a direct next step.

5. For bounded metrics such as work reduction, we report bounded bootstrap confidence intervals to respect logical constraints. With only five seeds, these intervals can still be wide in high-variability settings (e.g., GBM Covertype).
6. Statistical power. Five random seeds provide a limited basis for frequentist inference. All reported p-values should be regarded as rough indicators; non-significance should not be mistaken for evidence of no effect. Future work should increase the seed count (e.g., $\geq$20) to narrow confidence intervals and allow reliable effect-size estimation.
7. GBM convergence behavior is dataset-specific; Covertype shows high seed variability and weaker phase separation, suggesting that late boosting rounds can remain informative on heterogeneous multiclass boundaries.
8. Cross-model validation is currently limited to XGBoost. Replication on additional GBM implementations (e.g., LightGBM) and broader datasets is a direct next step.
9. QWYC ordering scope. Our QWYC-full implementation uses the greedy forward-selection ordering described by Wang et al. [6], specialized to the equal-weight RF setting: at each position, select the remaining tree that maximizes the count of margin-stopped validation instances. Wang et al.'s full framework additionally covers heterogeneous-cost classifier ensembles, a generalization that is not exercised here because standard RF trees have uniform cost and vote weight. For XGBoost, ordering optimization is inapplicable because boosted trees are sequential residual correctors.

9. Conclusions

We presented P^2-STOP, a family of nonparametric change-point stopping rules for sequential tree-ensemble inference. The default rolling-window variant provides an exact IQR scale with a finite-sample concentration guarantee (Theorem 1); the full-prefix P^2 variant offers $O(1)$-memory streaming at the cost of an approximate quantile whose convergence is empirically supported but not formally proved. Across four RF benchmarks and cross-model XGBoost validation, both variants substantially reduce tree-evaluation work relative to full-ensemble inference, with dataset-dependent accuracy trade-offs quantified by 95% confidence intervals and seed-level paired tests (5 seeds; inferential power is accordingly limited). Compared with QWYC [6]—evaluated in both margin-only and full ordering-optimized configurations—P^2-STOP offers complementary strengths: it achieves higher work reduction on datasets with gradual convergence (HIGGS, Covertype) while requiring no model-specific derivation.

The contamination stress test supports stable behavior of the IQR-scale functional with up to 25% random-tree corruption, consistent with its known 25% breakdown point; the composite stopping rule's formal breakdown point remains an open question. In the current Python prototype, work reduction does not fully translate into wall-clock speedup, motivating optimized low-level implementations, adaptive thresholding, and broader GBM/regression benchmarks. The Numba-compiled timing proof-of-concept with $T = 500$ indicates that speedup is consistently feasible on three datasets when work reduction is material, strengthening the practical case for an optimized deployment.

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